

Tools of the Trade: Linux and SQL

Activity: Filter with grep

Introduction

In this activity, I practiced using the `grep` command in Linux to search for specific information in files.

I also practiced using the pipe character (`|`) to pass the output of one command to another command.

The main command I practiced was:

- `grep`

I also used:

- `ls`
- `cd`
- Piping with `|`

These commands are useful when working with large amounts of information because they can help find specific information more quickly.

Activity Scenario

In this lab, I worked as a security analyst who needed to find specific information from server log files and user data files.

The tasks included:

1. Searching a server log for error messages.
2. Finding files with specific strings in their names.
3. Searching user files for specific usernames and department information.

The activity focused on using `grep` and piping to filter information from Linux files.

Lab Environment

- **Course:** Google Cybersecurity Certificate
 - **Course:** Tools of the Trade: Linux and SQL
 - **Activity:** Filter with grep
 - **Operating System:** Linux
 - **Shell:** Bash
 - **User:** analyst
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Task 1: Search for Error Messages in a Log File

The first task was to navigate to the logs directory.

I used:

```
cd logs
```

I then used the `grep` command to search the `server_logs.txt` file for lines containing the word `error`.

I used:

```
grep error server_logs.txt
```

The command returned only the lines that contained the `error` string.

After reviewing the results, I found that there were:

Six error entries

This showed me how `grep` can be used to quickly filter a log file and find specific types of messages.

The first argument in the command is the text being searched for, while the second argument is the file being searched.

Task 2: Find Files Containing Specific Strings

The second task was to search for files with specific strings in their names.

First, I navigated to the users directory:

```
cd /home/analyst/reports/users
```

I then used the `ls` command together with `grep`.

Find Files Containing `Q1`

I used:

```
ls | grep Q1
```

The `ls` command displayed the files in the directory, and the pipe (`|`) passed that output to `grep`.

The `grep Q1` command then filtered the output and displayed only the filenames containing `Q1`.

There were:

Three files

containing `Q1` in their names.

Find Files Containing `access`

Next, I searched for files containing the word `access` in their names.

I used:

```
ls | grep access
```

The command returned only the filenames containing the `access` string.

There were:

Four files

containing `access` in their names.

Task 3: Search File Contents

The third task was to search inside user files for specific information.

First, I displayed the files in the users directory:

```
ls
```

I then searched the `Q2_deleted_users.txt` file for the username `jhill`.

I used:

```
grep jhill Q2_deleted_users.txt
```

The search showed that the username:

jhill

was present in the file.

The answer was:

Yes, jhill was listed in the Q2 deleted users file.

Search for Human Resources Users

The next task was to find users who were added to the Human Resources department during quarter 4.

I searched the `Q4_added_users.txt` file using:

```
grep "Human Resources" Q4_added_users.txt
```

I used quotation marks around `"Human Resources"` because the search string contains two words.

The command returned the entries containing the Human Resources department.

The results showed that:

Two users

were added to the Human Resources department in quarter 4.

Understanding `grep`

The `grep` command is used to search for specific text inside files or filter command output.

For example:

```
grep error server_logs.txt
```

This searches the file for lines containing `error`.

Another example is:

```
ls | grep Q1
```

Here, the output from `ls` is passed to `grep`, which filters the output and shows only filenames containing `Q1`.

This makes it easier to find specific information without manually checking every line or filename.

Understanding Piping

The pipe character:

|

is used to send the output of one command to another command.

For example:

```
ls | grep Q1
```

The process works like this:

```
ls
↓
List files
↓
|
↓
grep Q1
↓
```

Show only matching files

Piping is useful because it allows multiple Linux commands to work together.

Commands Practiced

Command	Purpose
<code>cd</code>	Change to another directory
<code>ls</code>	List files and directories
<code>grep</code>	Search for specific text
<code> </code>	Pass output from one command to another

What I Learned

From this activity, I learned how to use `grep` to search for specific information in Linux files.

I practiced searching log files for error messages, finding filenames containing specific strings, and searching user files for usernames and department information.

I also learned how the pipe character can be used to connect commands and filter their output.

Cybersecurity Connection

Searching and filtering information is an important part of cybersecurity work.

Security analysts often need to examine logs and other files to find specific events, errors, usernames, or other information.

For example, a security analyst may need to search logs for:

- Errors
- Failed login attempts
- Usernames
- IP addresses
- Suspicious activity
- Specific keywords

Using `grep` can make this process faster because it allows an analyst to filter large amounts of text and focus on the information they need.

The ability to combine commands using pipes is also useful when working with Linux systems and security logs.

Practical Skills Demonstrated

- Linux Bash
 - `grep`
 - `ls`
 - `cd`
 - Linux file searching
 - Log filtering
 - Searching file contents
 - Searching filenames
 - Command piping
 - Basic log analysis
 - User data analysis
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Learning Outcome

This activity gave me practical experience with filtering information in Linux.

I learned how to use `grep` to search for specific strings in files and how to combine `ls` and `grep` using a pipe.

I also practiced searching server logs and user data files, which helped me understand how command-line tools can be used to find relevant information during security analysis.

Demonstration

The following screenshots demonstrate my hands-on work completed during this activity.

The screenshots show the Linux Bash terminal, the `grep` commands I used, the filtered results, and the searches performed on the server log and user files.

```

analyst@810fb420d9a2:~$ cd logs
analyst@810fb420d9a2:~/logs$ grep error server_logs.txt
2022-09-28 13:56:22 error    The password is incorrect
2022-09-28 15:56:22 error    The username is incorrect
2022-09-28 16:56:22 error    The password is incorrect
2022-09-29 13:56:22 error    An unexpected error occurred
2022-09-29 15:56:22 error    Unauthorized access
2022-09-29 16:56:22 error    Unauthorized access
analyst@810fb420d9a2:~/logs$ cd /home/analyst/reports/users
analyst@810fb420d9a2:~/reports/users$ ls | grep Q1
Q1_access.txt
Q1_added_users.txt
Q1_deleted_users.txt
analyst@810fb420d9a2:~/reports/users$ ls | grep access
Q1_access.txt
Q2_access.txt
Q3_access.txt
Q4_access.txt
analyst@810fb420d9a2:~/reports/users$ ls
Q1_access.txt      Q2_added_users.txt  Q3_deleted_users.txt
Q1_added_users.txt Q2_deleted_users.txt Q4_access.txt
Q1_deleted_users.txt Q3_access.txt       Q4_added_users.txt
Q2_access.txt      Q3_added_users.txt  Q4_deleted_users.txt
analyst@810fb420d9a2:~/reports/users$

```

```

analyst@810fb420d9a2:~/reports/users$ grep jhill Q2_deleted_users.txt
1025      jhill      Sales
analyst@810fb420d9a2:~/reports/users$ grep "Human Resources" Q4_added_users.txt
1151      sshah      Human Resources
1145      msosa      Human Resources
analyst@810fb420d9a2:~/reports/users$

```

The screenshots demonstrate my practical work with:

- grep
 - ls
 - cd
 - Piping with |
 - Searching server logs
 - Searching filenames
 - Searching user information
 - Filtering command output
-

Conclusion

This activity helped me understand how Linux commands can be used to search and filter information efficiently.

I practiced using `grep` to find error messages, search filenames, and locate specific information inside user files.

I also learned how piping can connect commands and make filtering information easier.

These skills are useful foundations for cybersecurity because security analysts often need to search through logs and system information to identify relevant events.

Course Information

Google Cybersecurity Certificate

Course: Tools of the Trade: Linux and SQL

Activity: Filter with `grep`

Disclaimer

This document is part of my personal cybersecurity learning portfolio. The activity was completed for educational purposes as part of the Google Cybersecurity Certificate.

The lab was performed in a provided Linux virtual environment, and the screenshots included in this document represent my practical lab work.